WEEK 4 · SUNDAY

Attention, GPT and speech

How chatbots read, think and talk.

AI for Beginners · Lesson 4, Part B (NLP)



TODAY'S PLAN

How a chatbot reads, thinks and talks

YESTERDAY IN ONE LINE

Words become **numbers**, and old models **forget** early words.

Today: the fix, and what it made possible.

1. **Attention**
   Looking at the right words, whatever the distance.
2. **GPT: built, trained, answering**
   The three lives of a language model.
3. **Kinds of models, and limits**
   Which model for which job, and where they fail.
4. **Hearing and speaking**
   Speech recognition (ASR) and text to speech (TTS).

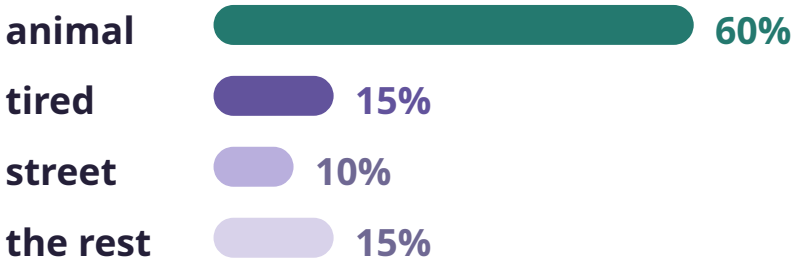
THE BIG IDEA

Attention: which words matter right now?

What does "it" mean?

The **animal** didn't cross the **street** because **it** was too **tired**

When reading "it", the model looks at:



Change one word: "...because it was too **wide**". Now "it" means the street.

Attention = each word decides how much to listen to every other word.

HOW ATTENTION WORKS

Like searching a library

Question

Each word asks: "what am I looking for?" The word "it" looks for a noun.

Label

Each word also carries a label: "what am I about?" "animal" says: a living noun.

Content

And the information it can hand over once someone picks it.

1. Match question and labels → **2. Turn scores into percentages** → **3. Mix the contents by those percentages**

Step 1 is similarity (Week 3). **Step 2** is softmax (Week 2). You already know the parts.

Remember the idea, not the jargon. (Experts call these query, key and value.)

THE TRANSFORMER

Stack attention in layers



Attention

Words share information with each other.

Think

A small neural network (Week 2) processes each word on its own.

Many layers

Early layers spot simple patterns, later ones grasp meaning.

Fixes the old problems: no forgetting, and all words are processed at once, so it is fast.

Born in 2017 in a paper called "Attention Is All You Need".

WHAT IS GPT?

Three words in the name

G

Generative

It creates new text, one piece at a time.

P

Pre-trained

It first learned from a huge amount of text, before you ever talk to it.

T

Transformer

The attention-based design we just saw.

ITS ONE JOB

The capital of Ethiopia is ___

Addis Ababa 92% | a 3% | the 2% | … 3%

Same game as yesterday's bigrams: predict the next token. The difference is that GPT uses **all** the earlier words, with attention.

ITS WORKING MEMORY

The context window

A long chat, as a strip of tokens



Counted in tokens

Your messages, its replies and any documents all use up the window.

No memory outside it

If it is not in the window, the model cannot use it. A new chat starts empty.

Growing fast

From a few thousand tokens to hundreds of thousands in recent models.

Practical tip: in a long chat, repeat the important facts. Do not assume it still remembers message 1.

HOW IT LEARNS

Three stages of training

1 Pre-training

Read a huge amount of text. Hide the next word, guess it, learn from mistakes.

Result: knows language and facts

2 Instruction tuning

Study thousands of example questions, each with a good answer written by people.

Result: answers instead of rambling

3 Feedback tuning

People rank two answers: which is better? The model learns what people prefer.

Result: helpful and safer replies

THE LOOP INSIDE guess → measure the loss → nudge the weights → repeat

You already know this loop. It is gradient descent from Week 2, just with billions of weights.

WHY "LARGE"?

From one weight to billions

Our Week 2 neuron

1

weight to learn

A large language model



...and so on, billions of times over

Data

Books, websites, code, articles. Trillions of tokens.

Parameters

The weights: the dials that training sets. Billions of them.

Computing power

Thousands of special chips (GPUs) running for weeks.

Cost matters: only a few organisations can train the biggest models. The rest of us use or fine-tune them.

WHEN YOU PRESS ENTER

One token at a time

Round 1 The capital of Ethiopia is → **Addis Ababa**

Round 2 The capital of Ethiopia is Addis Ababa → **.**

Round 3 The capital of Ethiopia is Addis Ababa. → **[stop]**

Read all the text so far → **Score every token** → **Pick one** → **Add it, repeat**

Training happens once. Answering happens every time you ask.

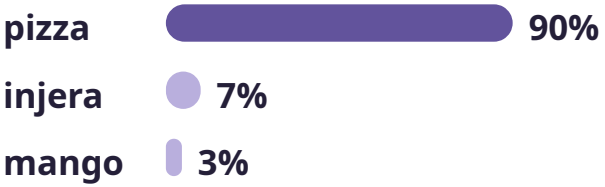
It never plans the full answer. It writes it as it goes.

HOW IT PICKS

Temperature: the creativity dial

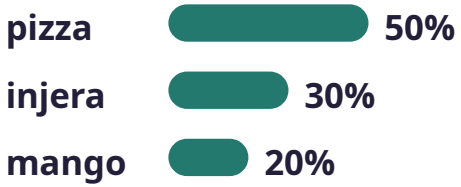
"My favourite food is ___"

Low temperature



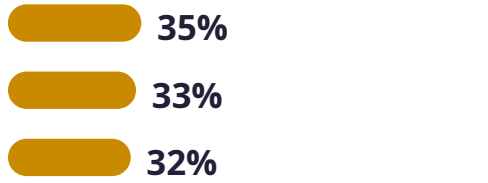
Safe. Same answer each time.

Medium



Balanced. Some variety.

High temperature



Surprising. Anything goes.

Low for facts, maths, code, summaries.

High for stories, names, brainstorming.

THE MODEL ZOO

Not every model is a chatbot

Readers

ENCODER MODELS

Understand text and turn it into numbers.

Used for: sentiment, search, spam filters.

Example: BERT

Writers

DECODER MODELS

Predict the next token, again and again.

Used for: chat, writing, code.

Example: GPT, Claude, Llama

Translators

ENCODER + DECODER

Read one text, write another from it.

Used for: translation, summaries.

Example: T5

Multimodal: text, images and audio together.

Small models: cheaper, can run on a laptop.

Open models: weights you can download.

Pick by the job: sorting 10,000 reviews does not need a giant chatbot. Writing a long email does.

USE WITH CARE

Where language models go wrong

Made-up facts
Sounds sure, but invents a book, a law or a quote.

Bias
Learns from human text, including its stereotypes.

Out of date
Only knows text up to its training cut-off date.

Language gaps
Weaker where little text exists online.

Three good habits

Check important facts. **Give it** the real documents to work from. **Never paste** passwords or private data.

Why it makes things up: it predicts likely words. Likely is not the same as true.

ASR · AUTOMATIC SPEECH RECOGNITION

How a machine hears



1. Sound wave

Thousands of numbers per second

→



2. Sound picture

Which pitches, when

→

3. Neural network

Attention again, now over sound

→

"Hello, how are you?"

4. Text

Words with punctuation

Hard cases: accents, background noise, fast talk, mixing languages.

You use it daily: voice typing, captions, voice notes to text.

TTS · TEXT TO SPEECH

How a machine speaks

1. Text
Dr. Amina has 3 cats

→

2. Say it out
Doctor Amina has three cats

→

3. Plan the sound
Pitch, speed, rhythm: the sound picture

→

4. Make the wave
A voice model turns the picture into audio

It is ASR in reverse: text to sound picture to sound, instead of sound to picture to text.

What makes it sound natural

Pauses, stress, rising tone for questions, and emotion.

Voice cloning risk

A few seconds of someone's voice can fake a call. Always ask consent.

PUTTING IT TOGETHER

A voice assistant, step by step

You speak
"Weather in Kigali?"

→

ASR
Sound to text

→

GPT
Understand and reply

→

TTS
Text to sound

→

You hear
"Sunny, 24 degrees"

Delay adds up

Each step takes time. A good assistant answers within about a second.

Newer trend

Some models handle speech directly, with no text in the middle.

What would you build? A voice helper in Amharic or Swahili? A reading buddy for younger kids?

RECAP

Four things to remember

Attention

Every word decides which other words matter. No more forgetting.

GPT = next token

One job, repeated: predict the next token, then add it and go again.

Train, then tune

Read lots of text, learn from examples, then learn from people's feedback.

Speech

ASR turns sound into text. TTS turns text into sound.

To do: finish the notebook, then take the n-gram and similarity quiz on Google Forms.

Next week: what is new in AI, and agents that take action.

BEFORE NEXT WEEK

Try it yourself

1. **"The trophy didn't fit in the suitcase because it was too big."** What is "it"? How does attention help?

2. A shop's chatbot **invents a return policy** that does not exist. Name the problem and one fix.

3. Low or high temperature? **(a)** summarise a medicine leaflet **(b)** ten names for a new café

Finish the notebook and quiz. Bring your answers next Saturday.